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Component of weight parallel to slope = $mg \sin \theta$

Component of magnetic force acting of rod in the opposite direction

$$= BIL \cos \theta = B \left(\frac{E_{\text{induced}}}{R} \right) L \cos \theta = B \left(\frac{BLv \cos \theta}{R} \right) L \cos \theta$$

$$\text{Thus, } mg \sin \theta = \frac{B^2 L^2 v \cos^2 \theta}{R}$$

$$\rightarrow v = \frac{mgR \tan \theta}{B^2 L^2 \cos \theta}$$

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